Bastion Password Manager

User Guide

Project Team (Group 36 — Toolkit)

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1 Introduction

Bastion Password Manager is a secure desktop application developed as part of the **Usable Security and Privacy** course project at IIIT Delhi. The system is designed to provide reliable credential storage, family sharing, and trusted-contact–based recovery while maintaining strong security guarantees and usability.

Key features include:

- End-to-end encrypted credential storage
- Local password generation and strength analysis
- Granular family sharing with domain-based restrictions
- Comprehensive audit logs for accountability
- Trusted-contact account recovery without revealing secrets

This guide explains installation, setup, everyday usage, sharing workflows, recovery procedures, security recommendations, and troubleshooting.

2 System Overview

Bastion consists of:

- **Desktop client** (Windows installer or Linux AppImage)
- **IIITD-hosted backend** for encrypted synchronization
- **Zero-knowledge architecture**: server never receives master passwords or decryption keys

Figure references in this guide correspond to interface screenshots.

3 Installation and Setup

3.1 System Requirements

- Windows (installer) or Linux (AppImage)
- Stable internet connection
- Access to the IIIT Delhi network for backend communication
- No macOS support in the current release

3.2 Windows Installation

1. Download `Group-36-Toolkit.exe`.
2. Run the installer and follow on-screen instructions.
3. Launch via Start Menu or desktop shortcut.

3.3 Linux Installation (AppImage)

1. Download `Group-36-Toolkit.AppImage`.
2. Make executable: `chmod +x Group-36-Toolkit.AppImage`
3. Execute: `./Group-36-Toolkit.AppImage`

3.4 Creating an Account

1. Click **Sign Up**.
2. Enter your email and master password.
3. Confirm to create your account.

Warning: The master password is never stored or transmitted; without trusted contacts, recovery is impossible.

4 User Interface Overview

This section briefly introduces the primary screens in the application.

4.1 Vault Home Screen

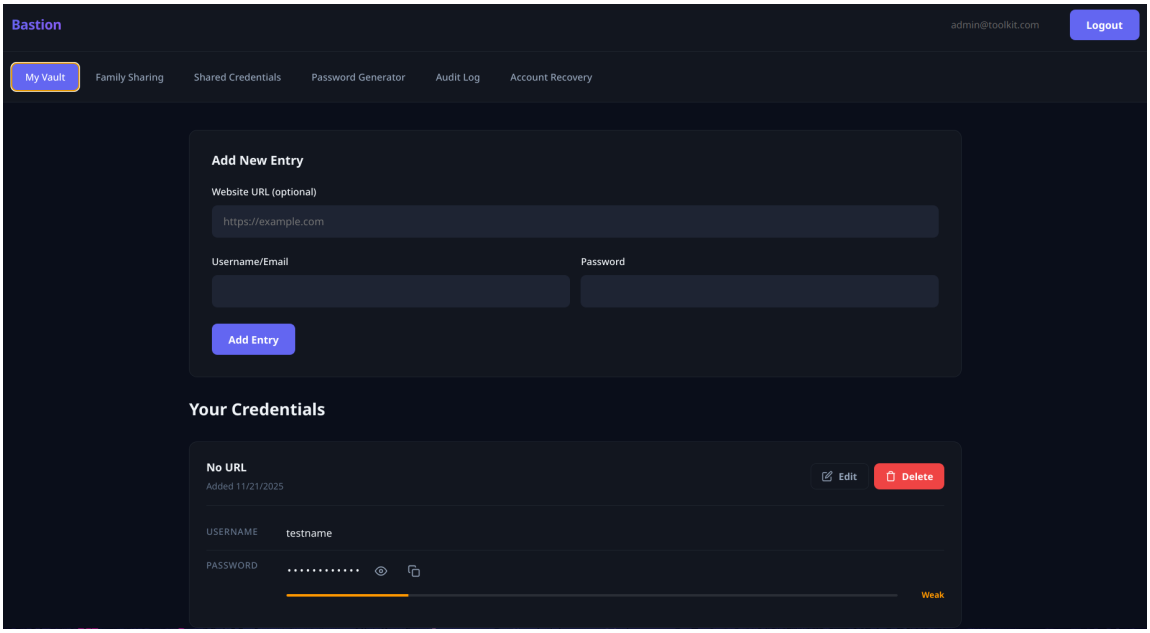

Figure 1: Add New Entry screen & Credential list view

The vault interface allows adding, viewing, editing, and deleting credentials.

5 Vault Management

5.1 Adding a Credential

Navigate to the Vault and fill in:

- Website URL
- Username/email
- Password (or generate one)
- Optional notes

Click **Add Entry**.

5.2 Editing or Deleting

- Edit: Select item → **Edit** → Save changes
- Delete: Select item → **Delete** → Confirm

Deletion is permanent.

5.3 Searching

Use the search bar to filter credentials by:

- Website name
- URL
- Username

6 Family Sharing

6.1 Overview

Family Sharing enables private credential usage across members without revealing plaintext passwords. Owners maintain full control.

6.2 Family Management

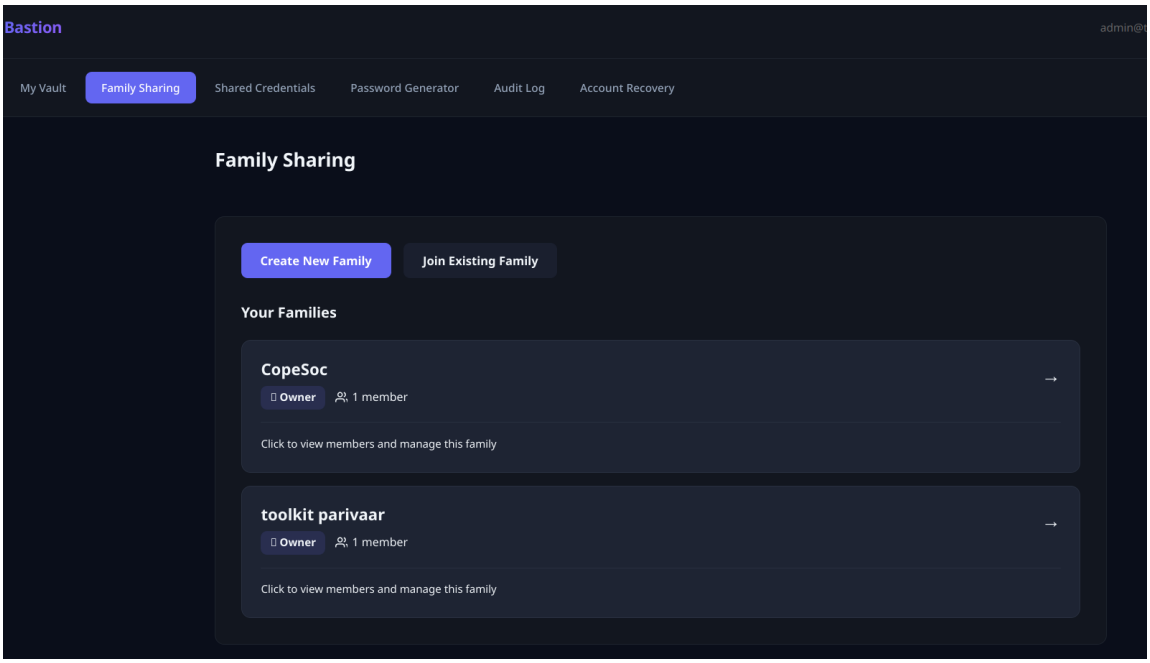

Figure 2: Family Sharing dashboard

6.2.1 Create a Family

Go to **Family Sharing** → **Create New Family**. Choose a family name.

6.2.2 Join a Family

Use an invite link generated by an owner.

6.3 Sharing Credentials

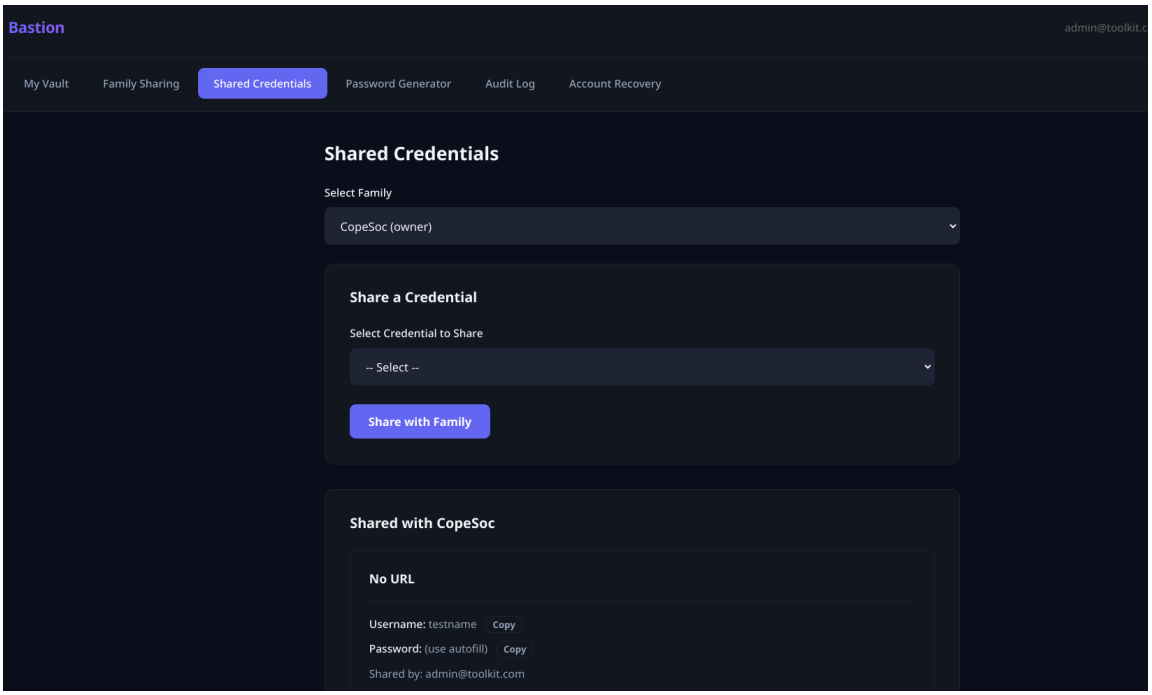

Figure 3: Shared Credentials interface

Steps:

1. Go to **Shared Credentials**.
2. Select a family.
3. Choose a credential to share.
4. Apply optional domain restrictions.
5. Click **Share with Family**.

Members may:

- Autofill
- Copy masked username

but cannot view plaintext passwords.

6.4 Revoking Access

Owners can revoke any share at any time; effect is immediate.

7 Password Generator

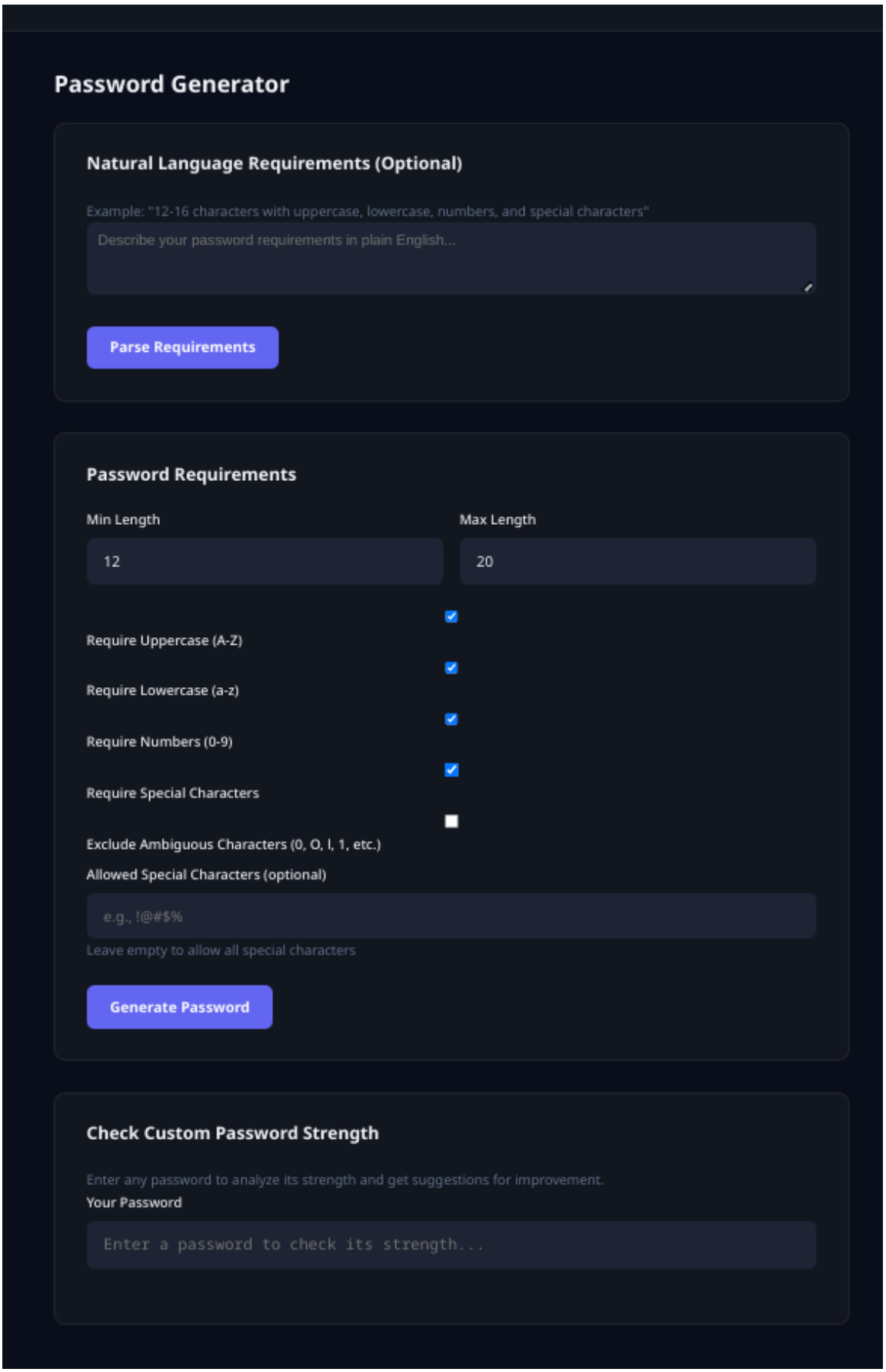

Figure 4: Password Generation screen

Features include:

- Customizable length (12–64)
- Character set options: uppercase, lowercase, numbers, symbols
- Ambiguous-character exclusion
- Strength meter
- Custom strength checks

8 Audit Log

The Audit Log provides visibility into shared-credential usage:

- Member
- Timestamp
- Domain
- Copy/autofill attempt
- Success/failure

This ensures accountability in shared families.

9 Account Recovery

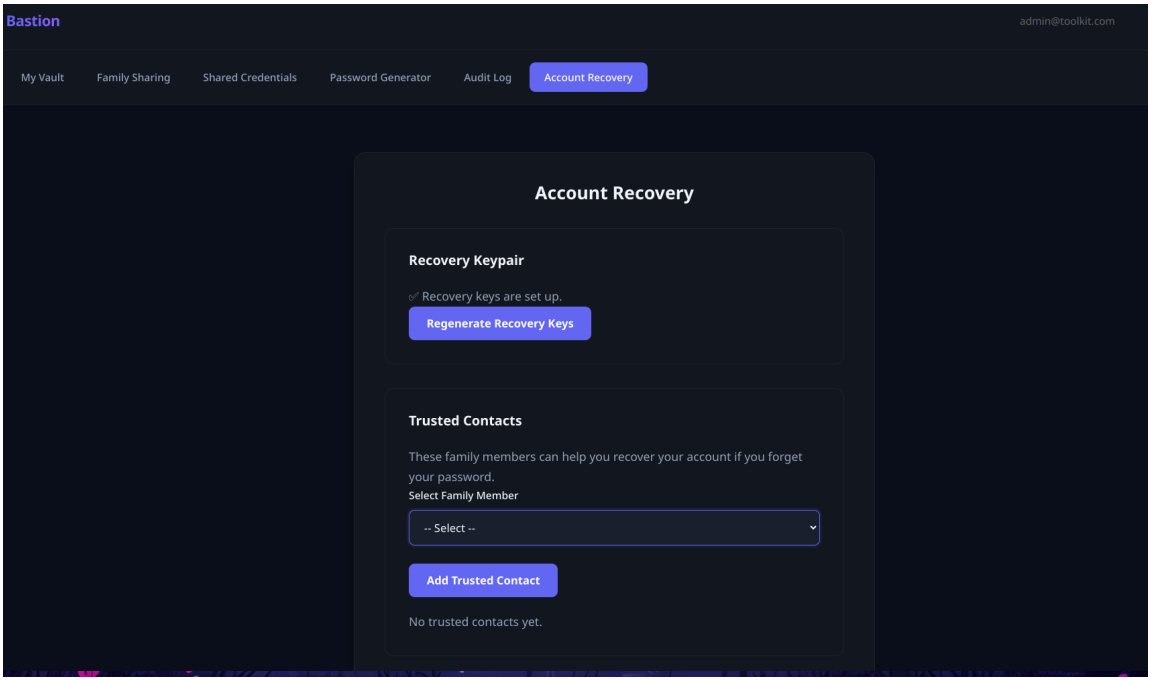

Figure 5: Account Recovery interface

9.1 Trusted Contacts Setup

1. Open **Account Recovery**.
2. Select family members as trusted contacts.
3. Add them to your recovery list.

9.2 Recovery Procedure

If you forget your master password:

1. Click **Forgot Password**.
2. Choose a trusted contact.
3. The contact authenticates and approves the reset.
4. You set a new master password.

Trusted contacts cannot access your vault.

10 Security Best Practices

- Use a strong master password (preferably a long passphrase).
- Do not reuse passwords—use the generator.
- Add 2–3 trusted contacts.
- Review audit logs regularly.
- Lock your vault when stepping away.
- Avoid using Bastion on untrusted/public systems.

11 Troubleshooting

Forgot Master Password

Use Trusted Contacts recovery. Without trusted contacts, data is unrecoverable.

Cannot Connect to Server

- Check internet and IIITD network access.
- Verify VPN and firewall settings.
- Contact system administrators if server downtime persists.

Shared Credential Missing

- Ensure invite acceptance.
- Refresh the vault or relogin.
- Verify domain restrictions.

App Fails to Start

Windows: reinstall or run as Administrator.
Linux: ensure AppImage is executable and FUSE is installed.

12 FAQ

Is my data safe?
Yes. All encryption happens locally; the server receives only ciphertext.

Can staff access my passwords?
No. Bastion follows a zero-knowledge design.

Does it work offline?
Read-only mode works after sync; editing requires connectivity.

What encryption is used?
XChaCha20-Poly1305 and Argon2id.

Appendix A — Security Architecture (Short Summary)

Local Encryption

All credentials are encrypted on the client using **XChaCha20-Poly1305**. Only ciphertext is synced to the backend.

Master Password Derivation

The master password is processed using **Argon2id** with high-memory parameters to resist GPU cracking.

Recovery Keypair

The app generates a local asymmetric keypair for recovery. Trusted contacts approve resets without accessing vault contents.

Family Sharing Model

Shared credentials use a secondary encryption layer. Members decrypt only metadata necessary for autofill—not plaintext passwords.

Zero-Knowledge Principle

Servers never store:
- master password
- derived keys
- recovery keys
- plaintext credentials

Contact & Support

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